

BE 1st Year, 2nd Test Nov 2023
Applied Physics 2APRC2 (ETC B and E&I)

Time: 70 min

Max. Marks: 20

All questions are compulsory. Attempt any one part from each question.

- Q1** I (a) Explain the construction, working and energy level diagram of CO₂ laser. What is the role of He in this laser? 7
- (b) What is V-number? If the core refractive index is 1.48 and a numerical aperture is 0.649, calculate the cladding refractive index. 2

OR

- II(a) Explain the absorption, spontaneous and stimulated emission of radiation. Obtain a relation between transition probabilities of spontaneous and stimulated emission. 7
- (b) What is optical resonator? A graded index fibre has a core diameter of 0.05mm and NA of 0.22 at a wavelength of 8500Å. What are the normalised frequency? 2
- Q2** I(a) Using the Maxwell's equations derive a wave expression for electric and magnetic field for vacuum. 7
- (b) State and explain the Ampere's law for steady current. Also show that it satisfies the continuity equation for steady current. 4

OR

- II(a) Derive the Maxwell's equation (i) $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ and (ii) $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$. 7
- (b) Derive an expression for the conservation of charge? 4
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